Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Online Information & Search System Mock Examination

Total Marks: *150 Marks*
Instruction: *prepared by Tr Jamal Fahadh for ICT candidates, Best academy, Hargeisa Somaliland*

Question 1: Classification & Mechanisms of Information Sources (Total: 35 marks)

Living in a connected world means that there is a lot of data available for individuals to find, though accessing high-quality assets demands methodical evaluation.

(a) State the definitions of a **primary source** and a **secondary source**. (2)

Primary Source:

Secondary Source:

(b) Identify four examples of **primary information sources** that an individual can create themselves. (4)

1. 2. 3. 4.

(c) Identify four examples of **secondary information sources** created by other people or organizations. (4)

1. 2. 3. 4.

(d) A student compiles a research matrix using external digital assets. Complete the table below by stating whether each item represents a primary or secondary information source based on the syllabus guidelines. (7)

Sourced Media or Information Item	Primary or Secondary Source Classification?
A podcast that you have created	*(i)*
A video of a discussion that you have with an expert	*(ii)*
Your own physical copy of a commercial film	*(iii)*
Notes that you take while visiting a local museum	*(iv)*
A printed book that you borrow from someone	*(v)*
An image file that you download from the internet	*(vi)*
Your own recording of a live radio broadcast	*(vii)*

(e) Describe how a standard **search engine** physically operates to locate and return data to an end-user. (4)

(f) Define the following internet search engine terminology parameters: (3)

- Keywords:

- Autofill:

- Browsing History:

(g) Analyze the famous technology quote by Mitchell Kapor regarding internet data collection mentioned in the text.

(i) State the exact analogy used in the quote. (1)

(ii) Explain the underlying information challenge this analogy represents for modern digital citizens. (2)

(h) Define the term **usage rights** regarding retrieved digital information pieces. (2)

(i) Identify one choice that correctly defines a primary info block. Place a cross (x) in the correct box. (1)

[ ] A. A textbook borrowed from a university library archive

[ ] B. An interview or questionnaire conducted entirely by you

[ ] C. A map downloaded from an online geography website

[ ] D. A television or radio broadcast captured by a network station

(j) State two ways in which an active image search can be refined by an application user. (2)

Question 2: Search Syntax Rules & Operational Applications (Total: 30 marks)

Search engine interfaces parse command operators to filter database records according to specific linguistic structures.

(a) Define what is meant by search **syntax**. (1)

(b) Complete the following search engine syntax rule and operational mapping table. (6)

Syntax Operator Symbol	Core Boolean logic / Operational Syntax Rule	Sourced Operational Output Impact
AND (+)	*(i)*	Returns only results matching both words.
NOT (-)	*(ii)*	*(iii)*
Speech Marks ("")	*(iv)*	*(v)*

(c) Identify one symbol utilized specifically to isolate a full, unbroken phrase with keywords kept in an exact specific order during a query execution. Place a cross (x) in the correct box. (1)

[ ] A. The plus symbol (+)

[ ] B. The hyphen/minus symbol (–)

[ ] C. Speech marks ("")

[ ] D. The commercial at symbol (@)

(d) A technology student executes three separate query string entries into a search engine. Predict the exact database filter action and result output generated by each syntax configuration. (6)

• laptop + charger:

• monitor - CRT:

• "quantum computing hardware":

(e) State three individual examples of search syntax operators covered within the specification text layout. (3)

(f) Explain why utilizing specific search syntax operators is more efficient than typing long natural language questions into an internet database search bar. (3)

(g) A user attempts to locate a stolen graphic design by typing query boundaries. Describe how combining a phrase match operator with a NOT operator can isolate specific stolen matches while excluding official agency portfolios. (4)

(h) State what a search engine does when an entered keyword has zero exact character matches inside its index files. (2)

Question 3: Evaluating Information Validity & Trust Metrics (Total: 35 marks)

Retrieved online data blocks must be systematically audited before they are accepted as facts or incorporated into active production workflows.

(a) Define the core metric known as **fitness for purpose**. (2)

(b) Complete the data verification roadmap below by entering the precise technical evaluation steps required to validate information. (10)

Validation Category	Precise Sourced Verification Mechanism / Checklist Activity
1. Accuracy Checks	• Verify information is from a trustworthy source with built-in fact-checking steps. • (i) • (ii)
2. Age Checks	• (iii)
3. Relevance Checks	• (iv) • Match the data's topic directly with the target topic of your project.
4. Reliability Checks	• (v)

(c) Analyze the parameters used to identify and expose intellectual imbalances or slanted data perspectives.

(i) Define the term **bias**. (2)

(ii) State two distinct checklist tasks an evaluator must perform to verify whether an online document is biased. (2)

(iii) State one critical reason why a researcher must proactively check whether data is biased before incorporating it into an official project. (1)

(d) Differentiate between an information piece's **relevance** and its **reliability** metrics. (4)

(e) Explain how comparing a suspicious claim against multiple unrelated external sources can successfully confirm its reliability tier. (3)

(f) Web logs and forums allow any anonymous user to publish text strings instantly. Explain why checking the publication date (Age Check) is critical when evaluating technical computing instructions found on a public web forum. (3)

(g) Explain how checking if an online article uses empirical evidence to back up its assertions can help verify its overall accuracy rating. (3)

(h) State two structural markers that suggest an online content source is highly trustworthy and handles data responsibly before hitting publish. (2)

Question 4: Plagiarism, Paraphrasing & Academic Integrity (Total: 25 marks)

The effortless copy-and-paste functionalities built into modern operating systems increase the prevalence of unverified data sharing and copyright violations.

(a) Define the terms **plagiarism** and **paraphrasing**. (2)

Plagiarism:

Paraphrasing:

(b) Describe the precise legal or ethical violation committed if a student copies another creator's intellectual assets into an assignment without stating that it is theirs. (2)

(c) Review the institutional consequences associated with academic dishonesty. State three distinct penalties or academic outcomes a student faces if they commit plagiarism within school, college, or university coursework. (3)

(d) Identify one primary defensive rewriting technique explicitly recommended in the textbook to avoid plagiarism claims. (1)

(e) State one specific benefit that students gain when they actively rewrite and reorganize research details instead of copying text word-for-word. (1)

(f) Explain why simply changing a few words in a copied paragraph without citing the original author is still considered a violation of intellectual integrity. (3)

(g) Explain why checking whether retrieved online documents are protected by copyright legislation is an essential step before embedding them into professional work channels. (3)

(h) Complete the following structural matrix regarding plagiarism and information reworking definitions. (10 marks, 5 marks per row)

Reworking Concept	Sourced Operational Mechanism / Core Activity	Sourced Target Purpose from Syllabus
Plagiarism	Copying someone else's original media files and pretending it is yours.	Violating rights; passing off work dishonestly.
Paraphrasing	*(i)*	*(ii)*

Question 5: Extended Response Essay (Total: 25 marks)

A regional academic research facility is launching an online library portal. The system allows students to aggregate cross-border historical documents, analyze media records, execute database keyword queries, and compile summarized research theses for peer-reviewed journals.

The facility's technology board is evaluating two information management models for users:
• **Option A:** A completely open, unmoderated retrieval layout that allows students to copy and paste external data segments freely, relies entirely on global algorithmic search engine indexes without syntax restrictions, and encourages bulk data gathering from alternative web logs and anonymous social platforms to maximize the volume of information.
• **Option B:** A highly structured, verified repository framework that requires users to pass all retrieved text blocks through automated paraphrasing evaluation tools, forces the use of strict search syntax queries (such as phrase matching, absolute exclusions, and explicit logical operators), and mandates triple-source cross-checking against independent, verified databases before any text can be exported.

Evaluate both options. You should discuss the impacts of each option on the factual accuracy and fitness for purpose of the research, the efficiency of filtering out irrelevant or biased data fields, the risk of students committing plagiarism or violating usage rights, and the overall research velocity for users tracking complex topics. (25)

Enter your extended evaluation response here...

End of Question Bank